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Friction-reducing and anti-wear properties of 3D hierarchical porous graphene/multi-walled carbon nanotube in castor oil under severe condition: Experimental investigation and mechanism study

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ABSTRACT

Traditional lubricants fail to meet the trend of green development. Consequently, eco-friendly hybrid nano lubricants are synthesized by incorporating 0.5 wt% 3D hierarchical porous graphene (3D HPG) and 0.7 wt% multi-walled carbon nanotubes (MWCNT) nanoparticles to castor oil. The performances of nano lubricants are determined by linear reciprocating wear test, and the obtained wear marks on steel balls and plates are evaluated by depth-of-field microscopy, scanning electron microscopy, energy dispersive spectrometer and Raman spectroscopy. The results of steel balls and steel plates wear analyses are highly consistent. Characterization results for steel plates show that the 3D HPG: MWCNT (0.5:0.7) achieves a reduction in friction coefficient and wear volume up to 26% and 48% at low speed and heavy load conditions (4 Hz - 20 N, 2.7 GPa), respectively. The mechanism of friction-reducing and anti-wear at low speed and heavy load is that 3D HPG is torn into 2D graphene nanosheets due to its easy shear performance, which forms a 2D graphene nanosheets/MWCNT/2D graphene nanosheets sandwich structure to be further involved in lubrication. This rolled bearing-like synergistic effect can further reduce friction and wear. All in all, the environmentally friendly nano lubricants consisting of 3D HPG/MWCNT hybrid nanoparticles and castor oil have sufficient potential for application in the field of heavy machine parts.

1. Introduction

According to statistics, an estimated 30% power consumption in the world is caused by wear and friction occurring at the rubbing surfaces [1,2]. Besides, there are more than 80% of mechanical part failures led by wear [3,4]. Therefore, it is extremely urgent to reduce the friction and wear existing in the mechanical component contact surfaces for saving energy and improving the reliability of products. Most commonly known solutions include the usage of additives and lubricants. However, when the mechanical equipment is subjected to more complex and severe working conditions, the wear resistance, friction reduction performances and consistency of general lubricants are still not to realize the lubrication requirements for mechanical devices. In order to solve this problem, there are a number of researchers who indicate that the properties of lubricating oil can be enhanced by adding nanoparticals, like IF-WS_2 [5], ZnO [6], SiO_2 [7], TiO_2 [8] and so on as additive. While the great progress that is made, nevertheless, whether a stable and effective lubricant film can be formed under harsh operating conditions remains to be further explored. Graphene, as a two dimensional nano material with single-atom thickness, possesses superior properties including high specific surface area [9], remarkable mechanical properties [10] and other excellent performances [11–13], which make it to be a potential candidate for lubrication field. Many reports show that graphene is added to solid and liquid-based lubricants in small amounts, this improves both anti-wear and anti-friction properties of lubricants drastically [14–19]. For example, it is perceived that adding graphene into water-based sand cleaning liquid can significantly decrease friction and wear of the CT80 coiled tubing [14]. In addition, it is also proved that the addition of graphene as additives of biodegradable oil accomplishes tool life increase and smoother surfaces simultaneously [16]. Moreover, graphene displays an excellent wear-resistance in the case of honed engine liner surfaces under high contact pressure [18]. These

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results verify that there are remarkable improvement in friction-reduction and wear-resistant behaviors of graphene as an additive to different lubricants in the field of tribology [19].

It is well known that structure determines performance, the tribological properties of graphene are largely influenced by its structure, in other words, signal-atomic layer graphene [20], multi-layer graphene [21–24] and modified graphene [25,26] have varying degrees of influence on their own tribological properties. Based on the tribological properties of graphene, researchers carry out further research around two dimensional graphene materials. It is proved that two dimensional graphene exhibits favorable friction-reduction and anti-wear effect either alone or mixed with other nanoparticles. Rasheed et al. concluded that the addition of 0.01 wt% few-layer 2D graphene (G12) as engine oil lubricant additives can effectively reduce friction and wear of piston rings [27]. Besides, Javeed et al. found that graphene oxide (GO) and detonated nanodiamond (DND) nanoparticle additives of different concentration compositions can significantly reduce the friction and wear of the original engine liner segments, especially for the combination of detonated nanodiamond and GO at a concentration of 0.5 mg/L, it achieves the greatest friction-reduction and anti-wear properties [28]. It was found by Mutyala et al. that a mixture of few-layer 2D graphene and MoS_2 as a solid lubricant significantly reduces friction and wear in steel and diamond-like-carbon (DLC) contacts at relatively high contact pressures (0.5 GPa–0.8 GPa) [29]. Since three dimensional graphene is composed of two dimensional graphene sheets stacked together, the lubricating properties of it should be further explored through experiments. Recently, the addition of a new type of 3D HPG is used to reinforce lubrication property, attracting widespread attention [30,31].

The 3D HPG has excellent mechanical properties such as compressibility, elasticity, etc. Besides, it also has the advantages of high yield and low cost [32,33]. In view of the above advantages and its unique structures, it is highly desirable to make further improvement on the tribological characteristics of the such graphene for researchers. Therefore, some scholars investigate its tribological performance as a lubricant additive. For instance, according to the previous study, 3D HPG with appropriate concentration, as a lithium grease additive, exhibited significant lubrication performance in three motion modes under heavy load conditions [31]. Qi et al. also found that the synergistic effect between 3D graphene and 2D hexagonal boron nitride can greatly improve the tribological properties of PAO 4, thus reducing friction and wear of steel/steel contact [34]. In addition, it was proposed by Gu et al. that the synergistic lubrication effect of 3D porous graphene and T161 (high molecular weight polyisobutylene succinimide) in PAO 6 dramatically decreases the friction and wear of 316 ASS surface under specific loads, which is attributed to the formation of a uniform lubricating protective film [35]. Nevertheless, there are still very limited reports on the tribological properties of 3D HPG. Meanwhile, it is also critical to select lubricants appropriately involved in the friction lubrication process. Compared to mineral oil, from the perspective of environmental protection, the eco-friendly vegetable oil is better suited to current economic trends owing to its superior biodegradability [36,37].

Previous reports demonstrate that such 3D HPG as a lubricant additive can exhibit excellent tribological properties. However, there are few reports on the synergy of such 3D HPG with other nanoparticles. Moreover, in a specific operating environment, such as low speed and heavy load, whether the 3D HPG can lubricate synergistically with other nanoparticles to achieve better tribological performance remains to be further explored. In addition, many researches on three-dimensional graphene are mainly based on mineral oil and synthetic oil. With the concept of green and sustainable development gaining popularity, the downsides of traditional lubricants become increasingly more visible, while the vegetable lubricants are increasingly valued for their natural advantages of economy and environmental friendliness. Besides, the vegetable lubricants can exhibit better friction-reduction properties. As one of the vegetable oils, the castor oil has biodegradability, high viscosity index and good lubricating properties relative to mineral oils, moreover, castor oil is an environment-friendly lubricant. These reasons make castor oil more suitable for low speed and heavy load applications. Therefore, 3D HPG and MWCNT nanoparticles, as castor oil additives, whose tribological properties and lubrication mechanisms deserve intensive exploration under low speed and heavy load conditions.

In this paper, to explore the tribological performances of 3D HPG, MWCNT and 3D HPG/MWCNT hybrid nanoparticles as castor oil additives under low speed and heavy load conditions, the tribological experiments are performed on a reciprocating tribometer under the contact pressure of 1.7 GPa–2.7 GPa and constant frequency of 4 Hz (48 mm/s), which is designed to simulate the working environment of large engineering equipment components, such as heavy load ball screw, etc. The worn surfaces are characterized successively by SEM, 3D depth-of-field digital microscopy and Raman spectroscopy. Afterwards, the EDS and contact angle are used to assist in exploring the formation mechanism of lubrication film, and it highlights the synergistic lubrication mechanism of 3D HPG/MWCNT hybrid nanoparticles. Compared to base oil, the results show that 3D HPG/MWCNT hybrid nanoparticles can significantly improve friction-reduction and anti-wear properties, where the 3D HPG:MWCNT (0.5:0.7) hybrid nanoparticles achieve the best tribological performances at heavy load. Based on the experimental results, it provides an effective tool to develop hybrid nanoparticles for the lubrication of heavy load machinery.

The main innovations of this study include the following points:

(1) An eco-friendly 3D HPG/MWCNT hybrid nanoparticle oil-based lubricant is developed.
(2) The 3D HPG:MWCNT (0.5:0.7) hybrid nanoparticle lubricant achieves the excellent friction-reduction and anti-wear effect under low speed and heavy load conditions.
(3) This paper reveals the mechanism of synergistic friction-reduction and anti-wear of 3D HPG and MWCNT hybrid nanoparticles based on experimental data and various characterization methods.

2. Specimen preparation and experimental process

2.1. Experimental materials

The 3D HPG nanoparticle is fabricated by using one-step ion-exchange and activation method, the acquisition process of graphene is based on two key considerations. Graphitization of the resin by local exchange of catalytic ions is one of the important factors. Moreover, adding heated KOH to the exchange resin to obtain high surface area 3D HPG materials is another one, and those procedures are also described in elsewhere [38]. The industrial grade MWCNT nanoparticles (with purity more than 90%, average inner diameter of 5 nm–10 nm and average outer diameter of 20 nm–40 nm, average length between 10 and 30 nm) are purchased from Shanghai Aladdin Bio-Chem Technology Co., Ltd (China). Castor oil used in the tribological tests is obtained from Xilong Scientific Co., Ltd (China), whose basic parameters are shown in Table 1. Both 3D HPG and MWCNT additives are six specified concentrations (0 wt%, 0.1 wt%, 0.3 wt%, 0.5 wt%, 0.7 wt%, 1 wt%), then, 0.5 wt% 3D HPG is mixed with MWCNT in diverse concentration ratios (0.5:0.3, 0.5:0.6, 0.5:0.7, 0.5:1) as additives to castor oil. The 3D HPG/MWCNT combinations with different mixing ratios are added to castor oil, shown in Table 2. All nanofluids are homogeneously dispersed via mechanical stirring at room temperature for 30 min. In accordance with previous

Table 1
The basic parameters of castor oil.

	20 °C Density (kg/m^3)	40 °C Kinematic viscosity (mm^2/s)	Flash point (close up) (°C)
Castor oil	0.950×10^3	232	229

Table 2
Constituent of nanofluids prepared in this study.

Code name	Constituent
Base oil	Castor oil
3D HPG	Castor oil +0.5 wt% 3D HPG
MWCNT	Castor oil +0.7 wt% MWCNT
3D HPG:MWCNT (0.5:0.3) nanofluids	Castor oil + 0.5 wt% 3D HPG + 0.3 wt% MWCNT
3D HPG:MWCNT (0.5:0.6) nanofluids	Castor oil + 0.5 wt% 3D HPG + 0.6 wt% MWCNT
3D HPG:MWCNT (0.5:0.7) nanofluids	Castor oil + 0.5 wt% 3D HPG + 0.7 wt% MWCNT
3D HPG:MWCNT (0.5:1) nanofluids	Castor oil + 0.5 wt% 3D HPG + 1 wt% MWCNT

literature and practical conditions, the duration of sonication is fixed at 30 min as the most desirable time [39,40]. There is no else dispersant being added to isolate the influences of nanoparticles additives. On top of this, the tribological samples, used in liner reciprocating friction tests, consist of steel plates (25 mm × 20 mm× 1 mm) and steel balls (3 mm diameter) made from SUS 304 stainless steel (210 HV, Ra 2 μm). Before the start of each experiment, the steel plates and balls are ultrasonically cleaned in an ethanol solution to obtain a clean surface.

2.2. Settlement experiments and tribological tests

2.2.1. Settlement experiments of nano additives in oil

Noticeably, the dispersion stability of nano additives in lubricants often limits their friction-reduction and wear-resistance properties. Thus, in this study, the pure castor oil containing 0.5 wt% 3D HPG, 0.7 wt% MWCNT and 3D HPG:MWCNT (0.5:0.7) is dispersed by sonication for 30 min and then put in different static vertical bottles for 17 days to observe the dispersion. By the static sedimentation experiments and ultraviolet–visible (UV) spectrophotometer, the dispersion stability of nano additives in castor oil can be evaluated.

2.2.2. Tribological tests of different loads

The linear reciprocating friction and wear tester (MFT-R4000) is applied to inquiry the tribological behavior of castor oil and the nanofluids at ambient temperature, whose ball-on-plate structure is composed of steel plate and steel ball, the steel ball is fixed against a sliding steel plate with a normal load. This linear reciprocating motion can be used to simulate the actual motion of heavy load ball screw. The specified schematic representation of the equipment is displayed in Fig. 1. The load applied to a stationary steel ball ranges from 5 N to 20 N that results in the corresponding contact pressure of 1.7 GPa–2.7 GPa. Besides, the rubbing velocity and duration of test are set at 48 mm/s (4 Hz) and 30 min, respectively. Therefore, all friction tests are in accordance with the working conditions of low speed and heavy load. In particular, there is a maximum contact pressure of 2.7 GPa when the applied load is 20 N. The stroke length is selected as 6 mm to facilitate the evaluation of the effect of friction and wear on the wear scars. All test parameters are given in Table 3. For the research idea of this paper, firstly, the optimum concentration ratios of additives (3D HPG and MWCNT, respectively) to the castor oil are achieved under 4 Hz - 10 N friction test conditions. Next, under the same test conditions, the optimal concentration ratio of 3D HPG and carbon MWCNT combination can be further investigated. Last but not least, to further assess the carrying capacity of the nanofluids under heavy loading conditions, the load ranges are extended from 5 N to 20 N. To generate reliable data, each friction test is repeated at least three times at room temperature for 30 min, the mean values of the obtained data are the final result.

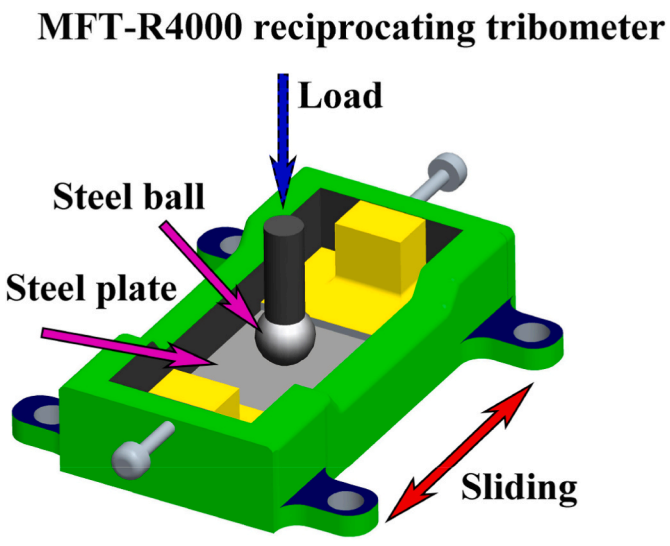


Fig. 1. Schematic representation of the reciprocating tribometer.

2.2.3. Maximum Hertzian contact pressure

The experiment is conducted under low speed and heavy load conditions, and the load is considered to be one of the critical factors affecting friction and wear, therefore, it is essential to calculate the maximum contact pressure between the steel plate and the steel ball surface. The formula adopted is as follows:

$$q^3 = \frac{6}{\pi^3} \times \frac{1}{R^2} \times \frac{P}{\left(\frac{1-\mu_1^2}{E_1} + \frac{1-\mu_2^2}{E_2}\right)^2} \quad (1)$$

where q represents the requested maximum Hertzian contact pressure; P represents the normal load, R represents the curvature radius of the steel ball; E_1 and E_2 represent the steel plate and steel ball modulus of elasticity, respectively; μ_1 and μ_2 represent the Poisson's ratio of steel plate and steel ball, separately; the parameters in the formula are all international standard units.

The value of parameter R is 1.5 mm, both μ_1 and μ_2 are 0.3; E_1 and E_2 are 194.02 GPa; the range of P is 5 N–20 N. Therefore, all values are put into the formula, the maximum Hertzian contact pressure q is calculated to be 1.7 GPa–2.7 GPa, which indicates that the experiment is always under heavy load conditions.

2.3. Characterization techniques

The micromorphology of the original 3D HPG and MWCNT is examined on SEM (Hitachi SU8220, Japan), furthermore, the SEM is also employed to further capture a clear microscopic morphology of the worn surface. Besides, for comparative analysis, Raman spectroscopy of both pristine nanoparticles and worn surfaces are obtained by Raman spectrometer (HORIBA, France). Meanwhile, the cross-sectional profile of scratch on the steel plate surface is detected by depth-of-field digital microscopy, which is also applied to estimate depth and width of the scratch profile, aiming at the wear loss of steel plate after the tribological experiment. Subsequently, it is the elemental distribution and content of the rubbing surface that are analyzed by EDS (BRUKER, USA). Finally, injecting a drop of 20 μL castor oil or nanofluids on the surface of the

Table 3
Parameters of the reciprocating friction test.

Condition	Values
Lubricating temperature (°C)	20–24
Friction load (N)	5–20
Hertzian contact pressure (GPa)	1.7–2.7
Test duration (minutes)	30
Reciprocating frequency (Hz)	4
Sliding speed (mm/s)	48
Stroke length (mm)	6

steel plate with a pipette, their wettability on the stainless steel plate is researched by optical contact angle measuring instrument (Theta Lite, Sweden), the magnitude of the contact angle is the average of the results obtained from at least three measurements.

3. Results and discussions

In order to clarify the final results, several necessary investigations are carried out in this chapter as follows: (I) Raman spectroscopy and SEM image characterization of pristine nanoparticle; (II) determination of the optimal concentrations of individual and blended nano additive; (III) comparative test of carrying capacity of mixed nano additive optimum concentration under heavy load conditions; (IV) analysis of SEM morphology and corresponding cross section profiles of rubbing surfaces; (V) exploration of Raman spectroscopy of worn surfaces; (VI) EDS analysis of worn surface and measurement of nanofluid contact angle; (VII) the proposal of a potential lubrication mechanism.

3.1. Characterizations of original 3D HPG and MWCNT

It can be clearly seen from Fig. 2(a), the image of 3D HPG reveals that the 3D HPG with a honeycomb coal framework structure is made by multi-layer graphene sheets stacked. As presented in Fig. 2(b), the SEM shows micrographs that MWCNT is tube-like and translucent, easily twisted and aggregated to form a network shape, which means that the intermolecular attraction is strong. Besides, its nanoscale size and tubular structure can make it adequately to fill surface pits. The Raman spectra of the pristine MWCNT are given in Fig. 2(d), the D (1343 cm^{-1}), G (1576 cm^{-1}) and 2D (2672 cm^{-1}) are the typical characteristic peaks of MWCNT, where the intensity ratio of D to G is about 1, as another isomer of carbon, Fig. 2(c) indicates that 3D HPG has similar characteristic peaks covering the D peak at 1342 cm^{-1}, G peak at 1570 cm^{-1} and 2D peak at 2676 cm^{-1}, the characteristics of small I_D/I_G (at 0.95) manifests that 3D HPG is relatively less defective. In addition, there is validated few-layer nature of pristine 3D HPG due to the appearance of 2D peak at high bands (2676 cm^{-1}), and the intensity ratio (I_G/I_{2D}) is much higher than 1.

3.2. Dispersibility of nano additives in castor oil

Fig. 3 shows the digital images of pure castor oil with 0.5 wt% 3D HPG, 0.7 wt% MWCNT and 3D HPG:MWCNT (0.5:0.7) at different standing times. The three nanoparticles solutions correspond to three bottles of the same specification for the standing precipitation experiment, the bottles are not disturbed and shaken during the experiment. To avoid reflections and to get a relatively clear and complete images, the walls of the same bottle look a little different on account of shooting angle.

As seen in Fig. 3 (a), the 0.5 wt% 3D HPG can be stably and homogeneously distributed in the pure castor oil, with no obvious precipitation and delamination after 17 days of resting. This 3D HPG, with a low number of oxygen-containing groups and non-polarity [38], possesses good hydrophobic and oleophylic properties, so it can be uniformly dispersed in castor oil and exist stably for a long period of time after ultrasonic dispersion. However, after being placed for the same time, the 0.7 wt% MWCNT shows obvious delamination and precipitation (displayed in Fig. 3(b)). The possible explanations for this phenomenon are that the MWCNT nanoparticles suspended in the upper layer of castor oil

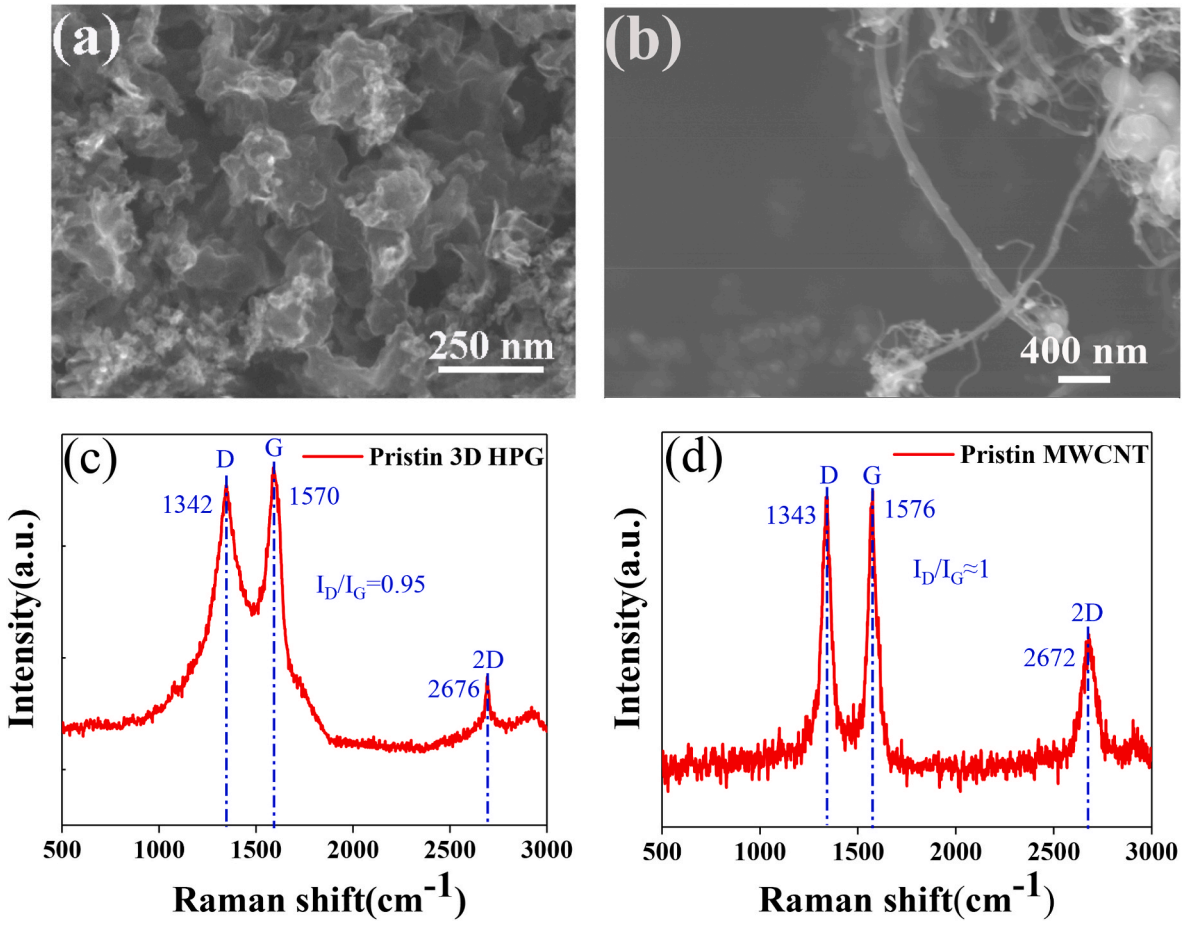


Fig. 2. Characterizations of pristine 3D HPG and pristine MWCNT. (a) 3D HPG SEM image; (b) MWCNT SEM image; (c) 3D HPG Raman spectrum; (d) MWCNT Raman spectrum.

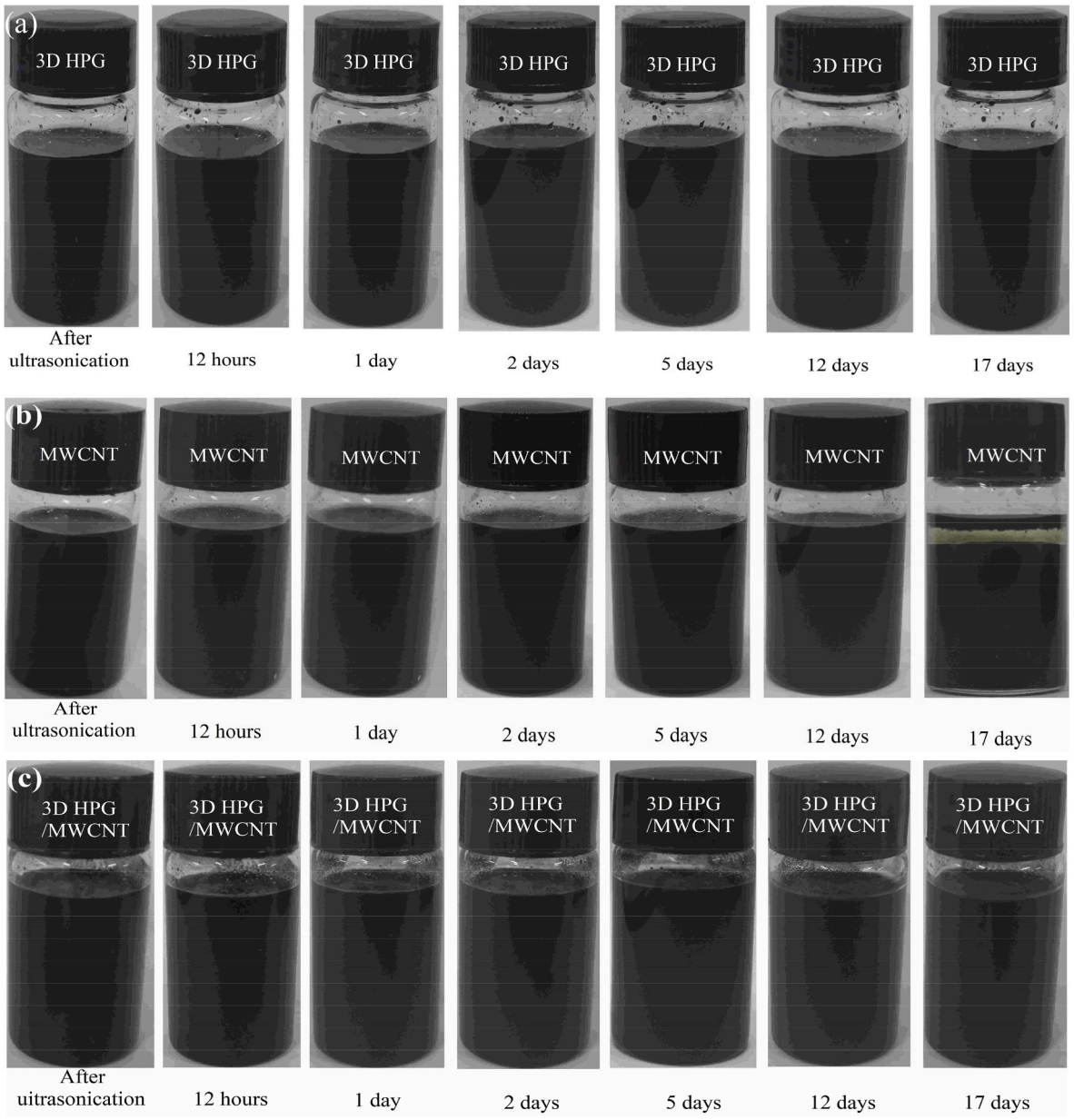


Fig. 3. Digital pictures of dispersibility at different times. (a) dispersibility of pure castor oil filled with 0.5 wt% 3D HPG; (b) dispersibility of pure castor oil filled with 0.7 wt% MWCNT; (c) dispersibility of pure castor oil filled with 3D HPG:MWCNT (0.5:0.7).

are prone to spontaneous aggregation and entanglement due to electrostatic forces and Van Der Waals forces. On the other hand, most of the MWCNT nanoparticles in the lower layer gradually precipitate because of their own gravity.

In Fig. 3(c) and D HPG:MWCNT(0.5:0.7) still maintains a uniform dispersion after 17 days. The delamination and reunion phenomenon are greatly improved when 0.5 wt% 3D HPG is added to pure castor oil containing 0.7 wt% MWCNT. Oleophylic 3D HPG, as another carbon material, plays a kind of glue role, the addition of it makes the three substances (pure castor oil, 3D HPG and MWCNT) to have good compatibility over a long period of time.

In addition, the dispersion stability of nanoparticles is quantified by the transmittance index of ultraviolet–visible spectrophotometer (in Fig. 4).

The transmittance of the nano lubricant means that the smaller the transmittance is, the better the dispersion stability of the nanofluid is. In Fig. 4(a), after ultrasonication treatment, the transmittance of the three nano lubricants are all less than 5%, indicating that the nano lubricants are well dispersed. With the increase of standing time, after 5 days (Fig. 4(b)), the transmittance of 0.7 wt% MWCNT nano lubricant increases to about 27%, while that of the 0.5 wt% 3D HPG and 3D HPG: MWCNT (0.5:0.7) nano lubricant is less than 20%. After 17 days (Fig. 4 (c)), the transmittance of the 0.5 wt% 3D HPG nano lubricant and 3D HPG:MWCNT (0.5:0.7) mixed nano lubricant is only about 25%, which is far lower than that of 0.7 wt% MWCNT.

In conclusion, the test results show that the 0.5 wt% 3D HPG and 3D HPG:MWCNT (0.5:0.7) nano lubricant have better dispersion stability in castor oil compared to that of 0.7 wt% MWCNT.

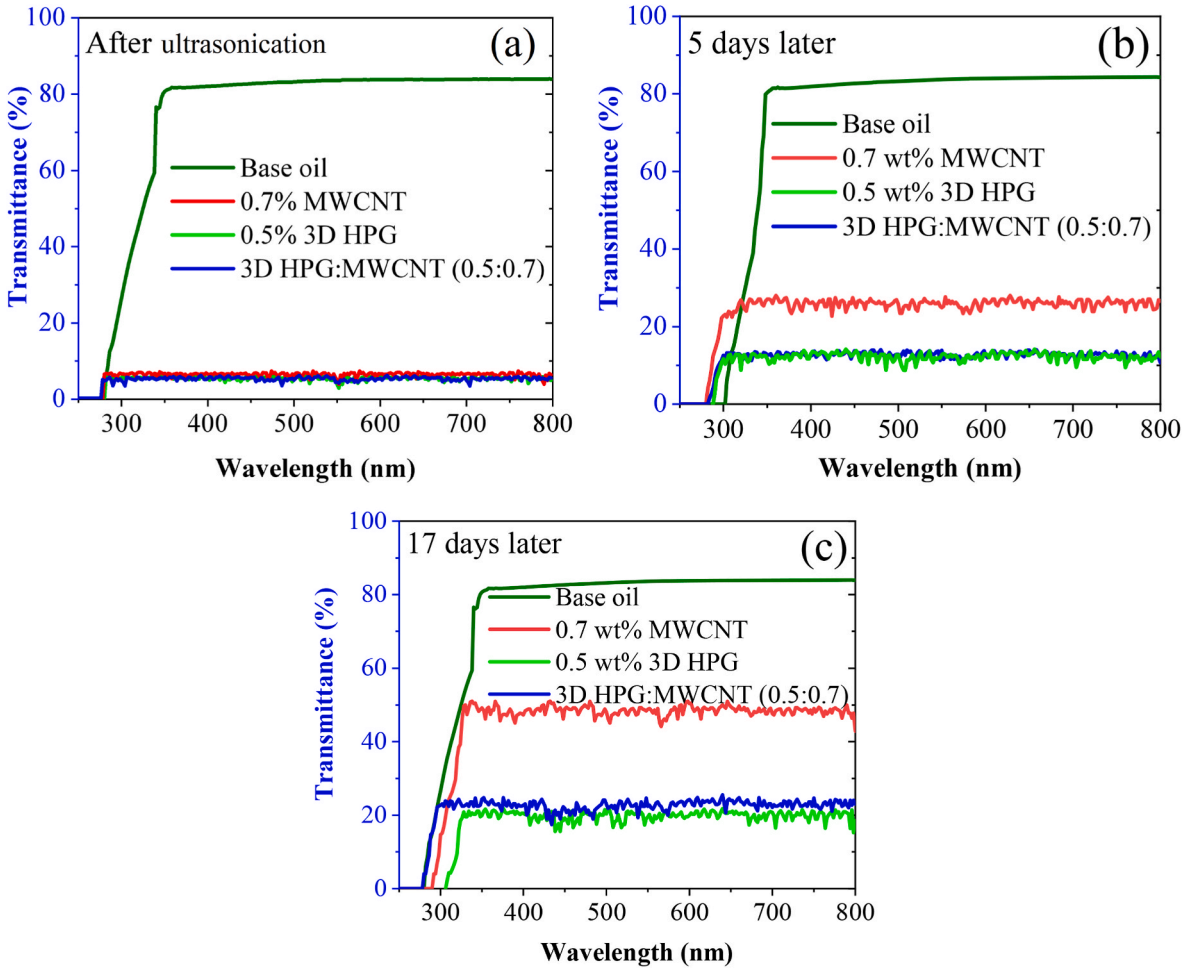


Fig. 4. Ultraviolet–visible transmittance after standing for different time. (a) after ultrasonication; (b) 5 days later; (c) 17 days later.

Table 4
Friction coefficient of 3D HPG and MWCNT at 0 wt%, 0.1 wt%, 0.3 wt%, 0.5 wt %, 0.7 wt%, 1 wt% cases (10 N, 4 Hz).

Parameter		10 N - 4 Hz					
		0 wt %	0.1 wt%	0.3 wt%	0.5 wt%	0.7 wt%	1 wt %
Friction coefficient (standard deviation)	3D HPG	0.053	0.048	0.046	0.044	0.048	0.049
		0.054	0.049	0.048	0.043	0.046	0.048
		0.052	0.047	0.044	0.046	0.050	0.051
	MWCNT		0.049	0.050	0.049	0.046	0.050
			0.052	0.051	0.051	0.047	0.048
			0.047	0.048	0.047	0.045	0.052

3.3. Friction and wear

3.3.1. Determination of optimal concentration

To reveal the tribological properties of 3D HPG and MWCNT as oil-based lubricant additives, the optimal concentrations for individual (3D HPG or MWCNT) and mixed (3D HPG/MWCNT) are investigated. The three test results of the friction coefficient are listed in Table 4 and Table 5. The idea of the friction experiment is that, firstly, the optimal concentrations of 3D HPG and MWCNT are measured separately, then the 3D HPG and MWCNT are prepared into blended solutions with the optimal concentration ratio. At the same time, several groups of 3D HPG/MWCNT blended solutions with different concentration ratios are set up as controls, where 3D HPG is always maintained at the optimal concentration. The lowest friction coefficient and wear rate of 3D HPG: MWCNT are measured, thus determining the optimal concentration

Table 5
Friction coefficient and wear volume data of 3D HPG and MWCNT at different concentration ratios case (10 N, 4 Hz).

Parameter	10 N - 4 Hz						
	3D HPG:MWCNT (wt:wt)						
	Base oil	0.5 wt% 3D HPG	0.7 wt% MWCNT	0.5:0.3	0.5:0.6	0.5:0.7	0.5:1.0
Friction coefficient (standard deviation)	0.053	0.044	0.046	0.045	0.042	0.038	0.042
	0.054	0.043	0.047	0.046	0.044	0.039	0.043
	0.052	0.046	0.045	0.044	0.041	0.040	0.045
Wear volume/mm^3	14233.1	9095.7	10172.5	9892.3	8606.2	7074.6	9205.3
	15034.3	8343.6	9469.8	10512.3	9209.5	6654.3	9795.6
	13423.5	9847.3	10875.4	9289.4	8004.6	7536.8	8615.3

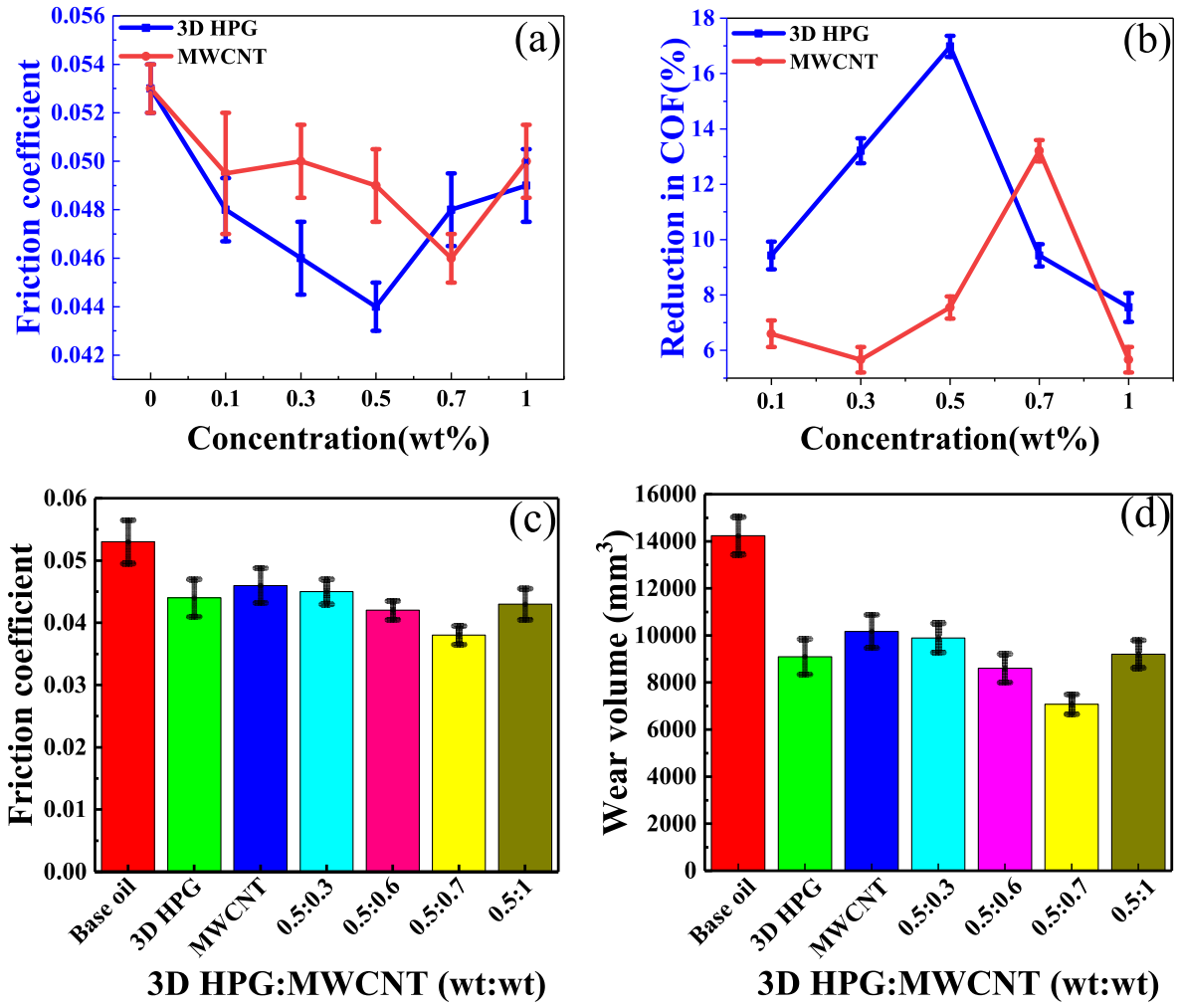


Fig. 5. Friction and wear of worn surface lubricated by different concentrations of 3D graphene, MWCNT and their mixing ratios (10 N, 4 Hz, 30 min). (a) and (c) average fiction coefficient; (b) the degree of friction coefficient reduction; (d) average wear loss.

ratio of the 3D HPG/MWCNT hybrid additive.

Fig. 5 discloses the test results of friction and wear under the experimental conditions of 4 Hz frequency, 10 N load and 30 min sliding time.

For the variation of average friction coefficient with nanoparticles additive content exposed in Fig. 5(a), the pure castor oil's friction coefficient is as high as 0.053, then, after adding the above two nano additives to the castor oil, all average friction coefficient values have different degrees of reduction, which indicates that both nano additives can play a role in friction-reduction. The overall trend of the average friction coefficient curves of the two nano additives is basically identical (decrease first and then increase), this means that there is an optimal concentration value corresponding to the minimum friction coefficient. For 3D HPG, the lowest value (≈0.044) is achieved at 0.5 wt% case, the reduction of friction coefficient is also greatest compared with base oil, approximately 17% displayed in Fig. 5(b). When the concentration of 3D HPG is below 0.5 wt% (0 wt% - 0.5 wt%), the appearance of higher friction coefficient (0.046–0.053) may be ascribed to the truth that too little 3D HPG can not form a uniform protective film between the friction pairs as a result of low density. However, exceeding this specific concentration, friction coefficient rises again, a plausible interpretation of this phenomenon is that excessive 3D HPG particles may be deposited to hinder the formation of lubricant film. Likewise, the optimal concentration of MWCNT added is around 0.7 wt%, which corresponds to a friction coefficient reduction of 13%. Besides, with a 0.7% concentration as a turning point, friction coefficient decreases and then increases, a reasonable explanation for this situation is that it is difficult to completely fill the metal surface defects when the MWCNT concentration is low, thus making it hard to form a stable lubricant film, when the concentration exceeds 0.7 wt%, the friction coefficient grows gradually, determined by the entanglement and agglomeration effect of MWCNT.

Fig. 5(c) shows the friction coefficient of 3D HPG/MWCNT mixed additives at different concentration ratios, the 3D HPG/MWCNT hybrid concentration exhibits smaller coefficients than that of the 3D HPG or MWCNT individually, which indicates that there is synergistic lubrication effect between 3D HPG and MWCNT in sliding process. Moreover, it is evident that the friction-reduction property of the 3D HPG/MWCNT combination is closely related to the mixing concentration ratio, when the MWCNT in the composite nanoparticle content increases from 0.3 wt % to 0.7 wt%, the friction coefficient reduces with the increasing of MWCNT nanoparticle content and then achieves the minimum value at 0.5:0.7, as it increases from 0.7 wt% to 1 wt%, the friction coefficient also goes up, the 3D HPG:MWCNT (0.5:0.7) shows the smallest friction coefficient (0.039) compared with that of 3D HPG (0.042) or MWCNT (0.044), respectively.

Fig. 5(d) shows a similar trend in the amount of wear, the experimental result of the base oil presents a fairly high wear volume, around 14233.05 mm^3, by contrast, when 0.5 wt% 3D HPG or 0.7 wt% MWCNT is added to the base oil, they produce wear loss of approximately 9095.73 mm^3 and 10172.48 mm^3, separately. However, it is still higher than the amount of wear generated by the four different mixed concentration combinations, likewise, the wear volume of 3D HPG:MWCNT

(0.5:0.7) falls to 7074.57 mm^3, which is nearly 51% lower than that of base oil.

In a word, under the working condition of 4 Hz–10 N, the tribological properties of castor oil can be improved in varying degrees by introducing 3D HPG, MWCNT and their four combinations of concentration ratios. Among them, the 3D HPG:MWCNT (0.5:0.7) as additives in castor oil has the best performance of friction-reduction and anti-wear.

3.3.2. Test of carrying capacity under heavy load condition

In view of the remarkable friction-reduction and anti-wear characteristics of 3D HPG:MWCNT (0.5:0.7) discussed above, it is essential to further evaluate the tribological behavior of 3D HPG:MWCNT (0.5:0.7) mixed concentration combinations under different degrees of heavy loads.

The three test results of the friction coefficient and wear volume are listed in Table 6. Fig. 6(a) and (c) illustrate the friction coefficient data for pure castor oil, 0.5 wt % 3D HPG, 0.7 wt% MWCNT, 3D HPG: MWCNT (0.5:0.7) at different loads.

In Fig. 6(a), the average friction coefficient of pure castor oil and the three different additives raise as the load gradually increases from 5 N to 20 N. Although all three additives can reduce friction, they do not have the same effect on friction-reduction at different loads. For instance, the average friction coefficients of pure castor oil, 0.7 wt% MWCNT, 0.5 wt % 3D HPG and 3D HPG:MWCNT (0.5:0.7) are 0.050, 0.040, 0.037 and 0.032 for 4 Hz - 5 N operating conditions, respectively. Compared to the pure base oil, the friction coefficients of the three different additives are reduced by 20%, 26% and 32%, separately. Besides, at 4 Hz - 20 N working condition, the friction coefficients of the four lubricants (0.061, 0.057, 0.054 and 0.045) correspond to a friction-reduction performance of 6%, 12% and 26%, respectively. It is worth noting that the friction coefficient of 3D HPG:MWCNT (0.5:0.7) decreases slightly when the load increases from 15 N to 20 N. The reason for this is that the viscosity and film thickness of castor oil decreases with the increase of load, which makes the suspended 3D HPG/HWCNT mixed nanoparticles can be fully adsorbed on the friction surface to play the role of friction-reduction. However, the individual nanoparticles (3D HPG or MWCNT) are easily squeezed out of contact area at heavy load, resulting in a continuous increase of the friction coefficient. These phenomena are also depicted in Fig. 6(c), with an increase in load, the friction-reduction efficiency of 0.5 wt% 3D HPG and 0.7 wt% MWCNT decline dramatically while 3D HPG:MWCNT (0.5:0.7) always has a favorable and uniform friction-reduction effect, this is more obvious in 4 Hz–20 N condition.

Fig. 6(b) and (d) show the variation of wear volume and wear-reduction rate for the four lubricants under different loads. In Fig. 6 (b), while the wear loss tends to increase with the increasing load, the differentiation of the anti-wear properties of the three nano additives becomes significant distinct. More specifically, under an applied load of 5 N, the reduction in wear volumes is 27.06%, 17.69%, and 33.1% for 0.5 wt% 3D HPG, 0.7 wt% MWCNT, and 3D HPG:MWCNT (0.5:0.7), respectively. Compared to the anti-wear performance of the three types of additives (28.58%, 23.09%, and 48.67%, separately) at 20 N load, all three nano additives have different degrees of wear resistance. However, the 3D HPG:MWCNT (0.5:0.7) reflects more outstanding anti-wear properties than the 0.5 wt% 3D HPG or 0.7 wt% MWCNT under heavy load (20 N) condition. These phenomena are validated by Fig. 6(d).

All in all, 0.5 wt% 3D HPG, 0.7 wt% MWCNT and 3D HPG:MWCNT (0.5:0.7) as oil-based additives can improve the tribological performance of pure castor oil at various loads. In addition, with a gradual increase in load (increase from 15 N to 20 N), 3D HPG:MWCNT (0.5:0.7) is more prone to maintain excellent friction-reduction and anti-wear properties than the other two nano additives. Moreover, in the case of relative heavy load, the 3D HPG:MWCNT (0.5:0.7) as an additive to pure castor oil realizes optimal tribological properties.

Table 6
The wear volume and friction coefficient lubricated by base oil; 0.5 wt% 3D HPG; 0.7 wt% MWCNT and 3D HPG:MWCNT (0.5:0.7) under different loads at 4 Hz.

Parameter		4 Hz			
		Base oil	0.5 wt% 3D HPG	0.7 wt% MWCNT	3D HPG: MWCNT (0.5:0.7)
Friction coefficient	5 N	0.005	0.037	0.039	0.032
		0.049	0.037	0.040	0.034
		0.051	0.040	0.042	0.035
	15	0.058	0.053	0.055	0.044
	N	0.054	0.052	0.052	0.046
		0.057	0.052	0.054	0.045
	20	0.059	0.055	0.057	0.043
	N	0.061	0.054	0.058	0.045
		0.062	0.054	0.058	0.044
Wear volume/ mm^3	5 N	10354.9	8522.8	7552.8	6927.4
		11306.5	9469.5	8517.4	7630.5
		9502.8	7527.6	6536.8	6224.8
	15	22126.8	13635.1	16549.2	11193.9
	N	20911.8	12722.6	15439.6	10549.7
		19696.4	145494	14353.3	11990.4
	20	29274.6	21618.5	22516.2	15026.6
	N	30397.7	20908.1	21080.5	16134.4
		27751.3	20192.8	23952.8	16453.8

3.4. Analysis of lubrication mechanism

3.4.1. Morphology of steel plate worn surface

To get a clearer visualization of the tribological performance of nano additives, Fig. 7 displays the SEM images and the cross-sectional profile of the worn surface measured by the 3D depth-of-field microscope under 4 Hz - 10 N working conditions.

Apparently, the abrasion width of pure castor oil is higher than that of the other three nano additives. In Figure (7), the width of the abrasion mark corresponding to pure castor oil is approximately 304.49 μm, compared with the case of 3D HPG:MWCNT(0.5:0.7) (about 183.29 μm), the abrasion width of the former is much wider. In addition, the depth of the abrasion mark also has a similar trend to the width. From the cross-sectional profile of the abrasion mark, it can be known that the castor oil has a maximum wear depth of 15.10 μm. After adding 0.5 wt% 3D HPG, 0.7 wt% MWCNT and 3D HPG:MWCNT (0.5:0.7) to castor oil, respectively, the width of wear mark decrease to varying degrees, of which the case of 3D HPG:MWCNT (0.5:0.7) achieves the lowest value of wear depth (around 9.73 μm). Meanwhile, according to Fig. 6(d), the corresponding reduction in wear is approximately 42.36% under this working conditions (4 Hz - 10 N).

When it comes to the morphologies of the worn surface, the surface lubricated by pure castor oil is rough and severely abraded (in Fig. 7(a)). Besides, there are a number of deep furrows and pits distributed on the worn surface, by contrast, the surface of the abrasion mark behaves relatively smoothly after using castor oil containing nano additives. Especially for castor oil containing 3D HPG:MWCNT (0.5:0.7) (in Fig. 7 (d)), whose wear surface has shallower furrows and fewer pits, moreover, its wear area becomes narrower. This means that the castor oil with 3D HPG:MWCNT (0.5:0.7) realizes the best anti-wear effect and the smoothest wear surface among the four lubricants.

In short, compared to individual nanoparticle additives (0.5 wt% 3D HPG or 0.7 wt% MWCNT), the castor oil with 3D HPG:MWCNT (0.5:0.7) can significantly minimise abrasive and adhesive wear and thus further enhance the tribological performance of castor oil.

3.4.2. Morphology of steel ball worn surfaces

The steel ball is the counter material in the friction pair, therefore, the wear analysis is conducted on it. Fig. 8 presents the wear of the steel ball surface detected by the depth-of-field microscope.

The wear surface morphology of steel ball under different lubrication conditions is shown in Fig. 8(aI)-(dI). From Fig. 8(aI) to Figure (dI), the

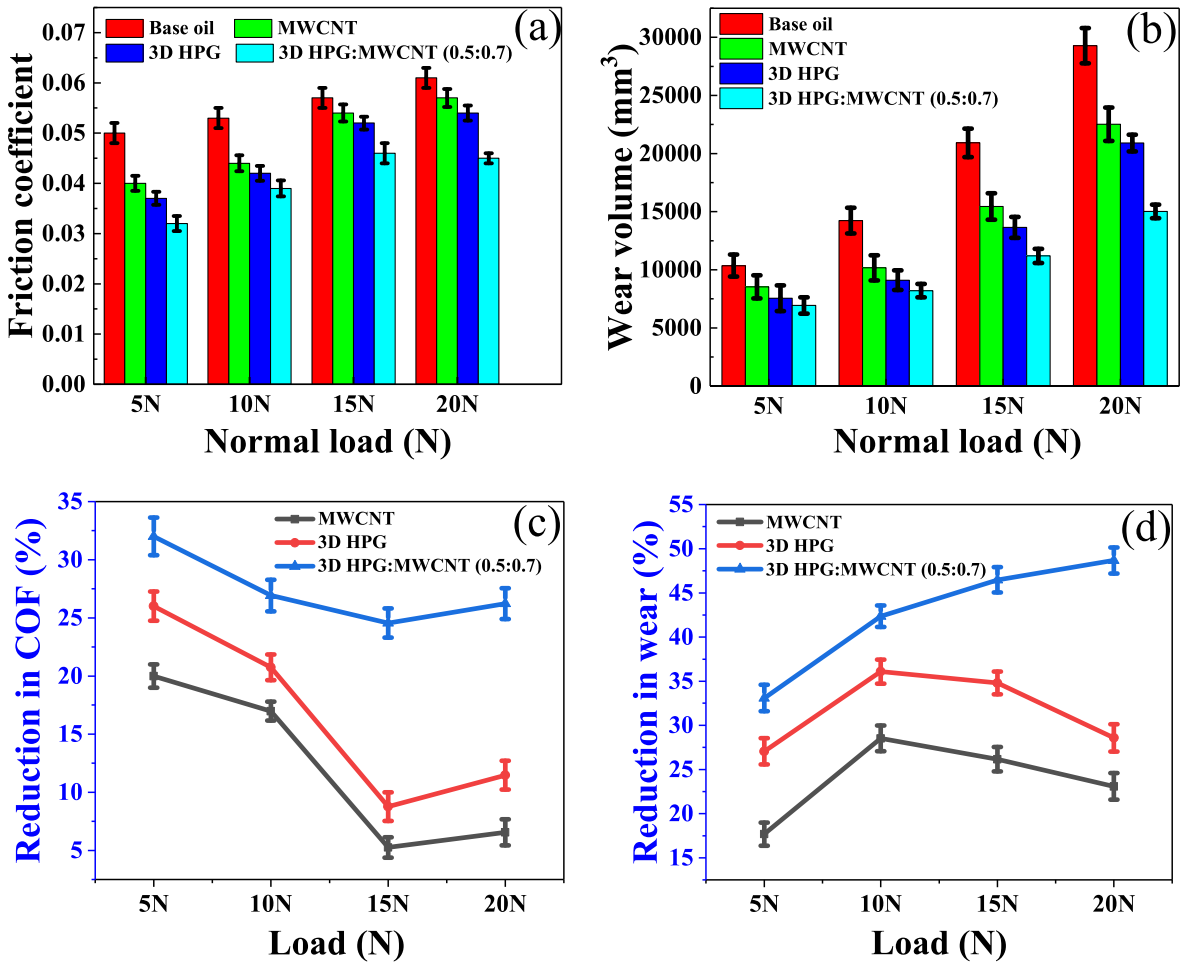


Fig. 6. Friction and wear of lubricated surface under variable load conditions (4 Hz, 30 min). (a) and (b) the average friction coefficient and the average wear volume; (c) and (d) the reduction in friction coefficient and wear volume at four different loads.

wear surface becomes progressively smoother, beside, for Fig. 8(aI), dense grooves and vertical scratches scatter over the surface lubricated by the base oil, which is mainly caused by the adhesive wear and abrasive wear between the steel-steel friction pairs. In the case of pure base oil lubrication with high viscosity, adhesive wear easily occurs within frictional pairs. In the subsequent sliding process, the groove-like abrasion marks on the surface of the steel plate are generated due to the furrowing action of the flaked debris and abrasive particles. Meanwhile, the surface migration of the material is completed, which can be verified in Fig. 7(a). The depth of grooves and scratches gradually becomes shallower after adding the nanoparticles (shown in Fig. 8(bI), (cI), (dI)), especially for the case of castor oil containing 3D HPG:MWCNT (0.5:0.7). This also means that the effects of abrasive and adhesive wear are dramatically reduced, that is to say, it achieves the best lubrication effect. In addition, from the cross-sectional profile presented in Fig. 8 (aIII)-(dIII), compared with the base oil, the width and depth of the wear marks lubricated by the 3D HPG:MWCNT (0.5:0.7) decrease significantly, thus, greatly reducing the wear volume (the blue dashed line represents the fitted profile of the original steel ball). The three-dimensional morphology of the worn steel ball also confirms this view (Fig. 8(aII)-(dII)).

In summary, the above analysis results show that the conclusions of the wear analysis of steel balls are basically the same as those of steel plates, and the wear scars on steel plates are somewhat intuitive. Therefore, for convenience, only the wear marks on the steel plate will be analyzed in the subsequent sections.

3.4.3. *Raman spectrum of friction surface*

In order to understand the active role and lubrication mechanism of nanoparticles, Raman spectra of the four lubricants worn surfaces (4 Hz - 10 N conditions) are specifically analyzed, as indicated in Fig. 9.

Obviously, the peak at about 690 cm^{-1} (appears in Fig. 9(a)–(d)) is the characteristic peak of iron oxide, this indicates that the sliding contact between the steel ball and the steel sheet cannot be completely avoided with or without nanoparticles lubrication, and the tribological experiments are in the case of boundary lubrication. For surfaces lubricated by base oil in Fig. 9(a), no other peaks are found except for the signature peak of iron oxide. However, the D peak (around 1343 cm^{-1}), G peak (around 1576 cm^{-1}) and 2D peak (around 1343 cm-1) of MWCNT are presented in Fig. 9(b), suggesting that MWCNT is involved in the friction lubrication process, in comparison to the original MWCNT (depicted in Fig. 2(d)). The intensity ratio of the D peak to the G peak (I_D/I_G) increases (from ≈1 to > 1), which reflects the increased structural damage and defect of the MWCNT after completing friction test, besides, more prominent D+D' peak indicates that MWCNT nanoparticles enter the friction interface to help with lubrication. Similarly, like Fig. 9(c), compared with Fig. 2(c), a substantial increase in I_D/I_G (from about 0.95 to over 1) suggests that the disorder and defect density of 3D HPG structure grows, which is attributed to the tearing action and the graphitization of the 3D HPG under heavy load. Moreover, the decline in G peak intensity indicates a reduction in the number of 3D HPG layers, which may be due to the following reasons that the complete three-dimensional layer structure of 3D HPG is torn into two dimensional sheets during the frictional process.

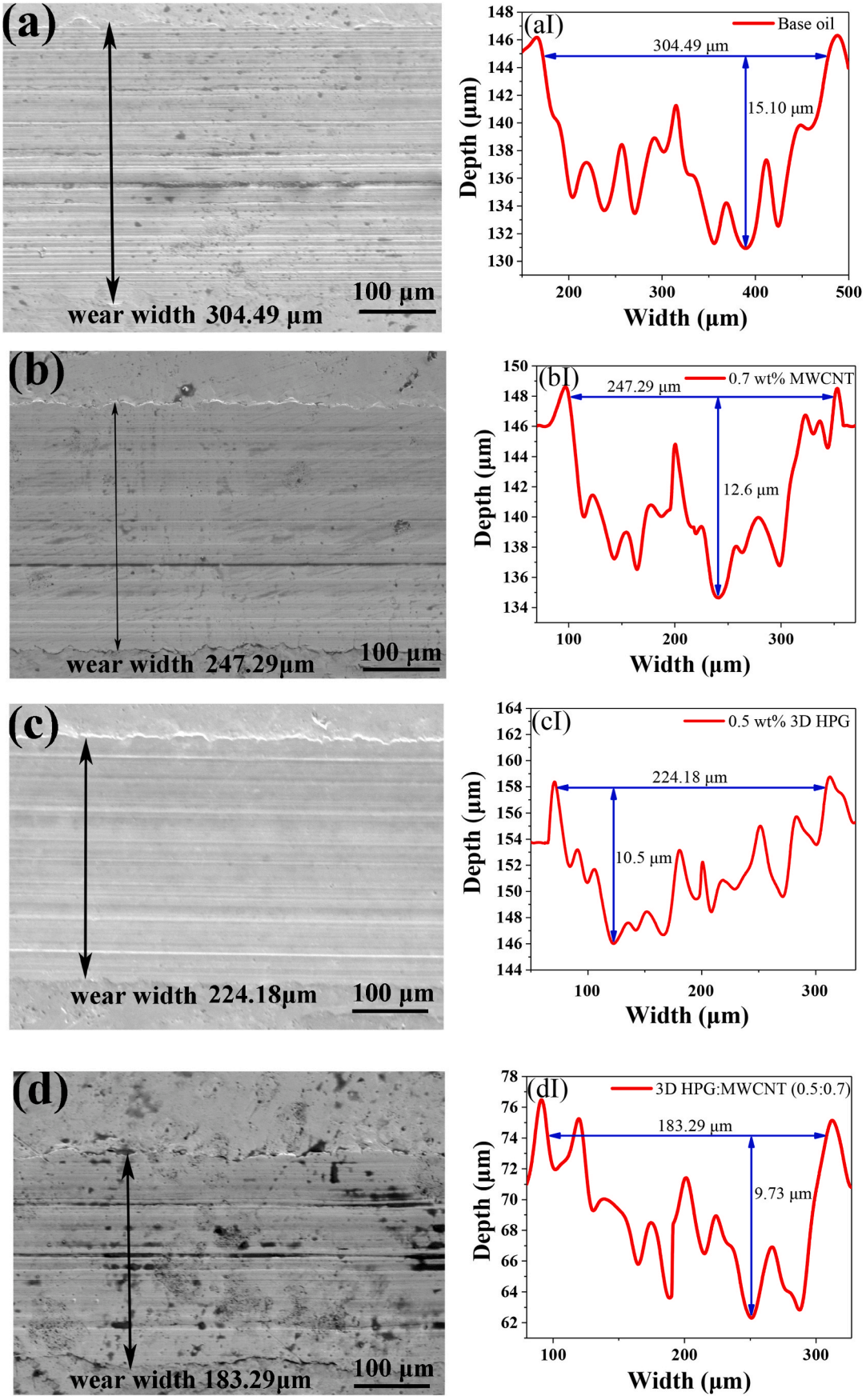


Fig. 7. SEM morphology of the worn surfaces and corresponding cross section profiles lubricated by different nanofluids on 304 stainless alloy steel under 4 Hz - 10 N working conditions. (a) and (aI) base oil case; (b) and (bI) 0.7 wt% MWCNT case; (c) and (cI) 0.5 wt% 3D HPG case; (d) and (dI) 3D HPG:MWCNT (0.5:0.7) case.

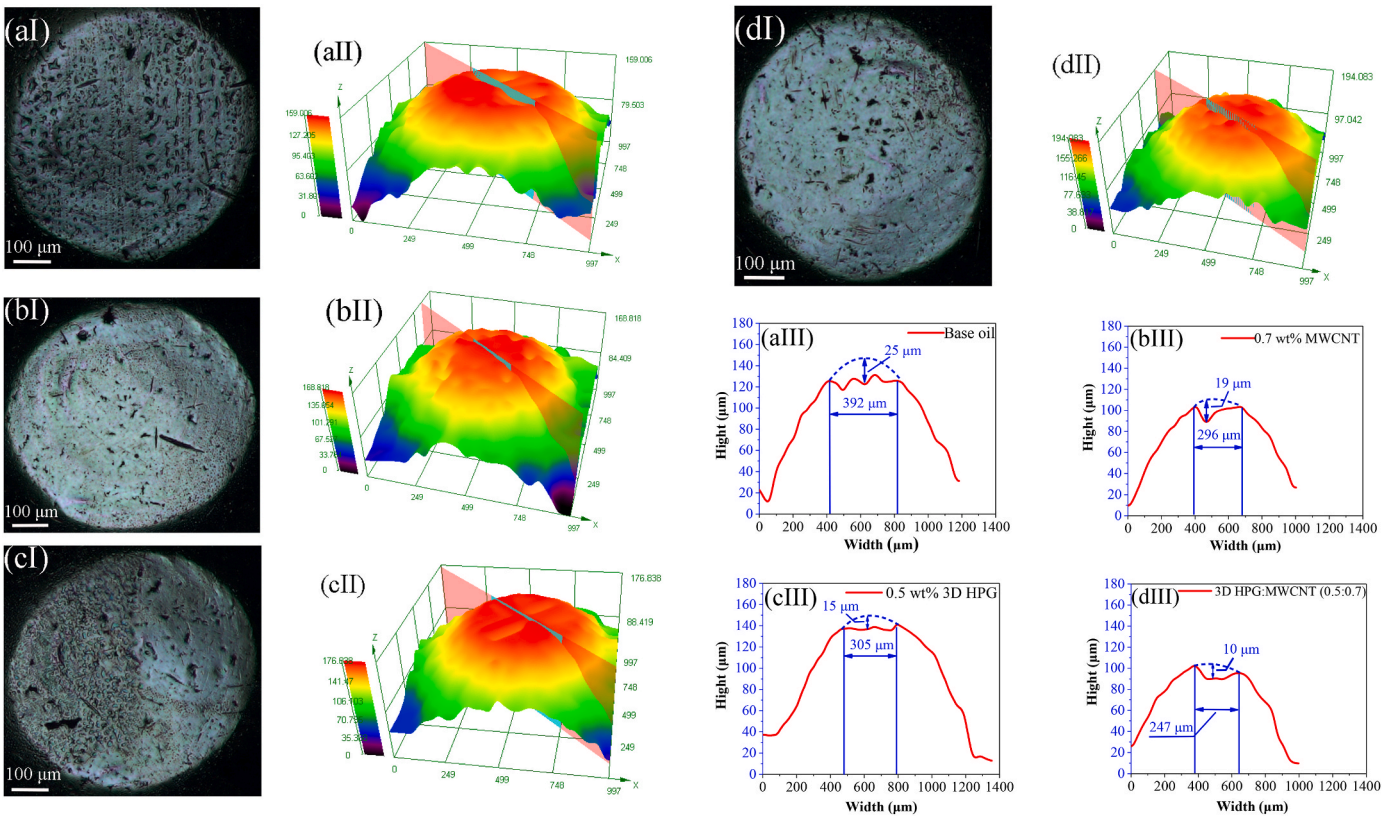


Fig. 8. The micromorphology, three-dimensional image and cross-sectional profile of the wear surface of steel balls under 4 Hz - 10 N working conditions. (aI)-(aIII) base oil; (bI)-(bIII) 0.7 wt% MWCNT; (cI)-(cIII) 0.5 wt% 3D HPG; (dI)-(dIII) 3D HPG:MWCNT (0.5:0.7).

For Raman spectra of worn surfaces lubricated by 3D HPG/MWCNT hybrid nano additives, as depicted in Fig. 9(d), the intensity ratio of the D peak to the G peak ($I_D/I_G \approx 1$) is smaller, compared to that of the individual MWCNT (in Fig. 9(b), $I_D/I_G > 1$) or 3D HPG (in Fig. 9(c), $I_D/I_G > 1$) nano additives, this means that the hybrid nanoparticles form a stable organizational structure, hence, its structural disruption and disorder become smaller under high contact pressure. Furthermore, the intensity of the G peak of 3D HPG/MWCNT hybrid nanoparticles (in Fig. 9(d)) is higher than that of 3D HPG (Fig. 9(c)), but similar to that of MWCNT (Fig. 9(b)). The possible reasons for these results are that the intensity of the G peak is roughly positively correlated with the number of layers of carbon material (within 10 layers), compared to the individual 3D HPG or MWCNT nanoparticles, that 3D HPG/MWCNT hybrid nanoparticles are subjected to uniform stress under the heavy load, therefore, after the friction process, the 3D HPG and MWCNT layers structures in hybrid nanoparticles are relatively well preserved.

By the above analysis of Raman spectra, we may conclude that there is synergistic lubrication effect between 3D HPG and MWCNT nanoparticles in the 3D HPG/MWCNT hybrid nanoparticles, and this synergy effect enables each other to maintain relatively good structural integrity. Meanwhile, due to this synergistic lubrication effect, compared with 3D HPG or MWCNT nanoparticles alone, the 3D HPG/MWCNT hybrid nanoparticles can further improve the tribological performance of base oil.

3.4.4. *EDS analysis of rubbing surface and nanofluid contact angle*

To further confirm the positive effect of the nanoparticle additives, the EDS data of the worn surfaces lubricated by the base oil and the nanoparticle additives is shown in Fig. 10. Correspondingly, Fig. 11 shows the contact angles of the base oil and three different nanofluids on the surfaces of SUS 304 stainless steel.

In Fig. 10(a), there is a small amount of carbon and oxygen element on the worn surfaces lubricated by the base oil, attributed to the partial oxidation of the wear surface of the stainless steel sheet during the friction process. The elements iron and chromium originate from the base components of the steel-steel tribological pair. By contrast, the content of element carbon increase significantly at 0.5 wt% 3D HPG (in Fig. 10(b)), 0.7 wt% MWCNT (in Fig. 10(c)) and 3D HPG:MWCNT (0.5:0.7) (in Fig. 10(d)) cases, which is strong evidence that the nanoparticles have adsorbed on the worn surface and formed a friction protection film. This phenomenon suggests that nanoparticles can play an enhanced lubricating role at the friction interface, in accordance with the experimental results of friction-reduction and anti-wear performances.

For the wettability of nanofluids on steel alloy surfaces, given in Fig. 11(a)–(d), the contact angles of base oil, 0.7 wt% MWCNT nanofluids, 0.5 wt% 3D HPG nanofluids and 3D HPG:MWCNT (0.5:0.7) composite nanofluids are 51°, 41°, 39° and 30°, separately. The better the wettability (meaning the smaller contact angle) is, the easier it is to form a lubrication film on the surface of friction pair. Compared to the base oil case, the maximum reduction in contact angle of the 3D HPG:MWCNT (0.5:0.7) composite nanofluids is about 41%. Therefore, the 3D HPG:MWCNT (0.5:0.7) composite nanofluids is much easier to form a protective film on the surface of friction pair to play a role in friction-reduction and anti-wear.

3.5. *Lubricant mechanisms*

The above discussion analysis clearly indicates that the overall tribological properties of the 3D HPG:MWCNT (0.5:0.7) blend lubricant is superior to that of the base oil, pure 3D HPG and pure MWCNT lubricants. This outstanding performance is ascribed to the synergistic lubrication effect of 3D HPG nanoparticle and MWCNT nanoparticle suspended in the base oil. Fig. 12 gives the lubrication mechanism when using pure 3D HPG nanoparticle, pure MWCNT nanoparticle and 3D HPG:MWCNT (0.5:0.7) hybrid nanomaterials as oil-based additives.

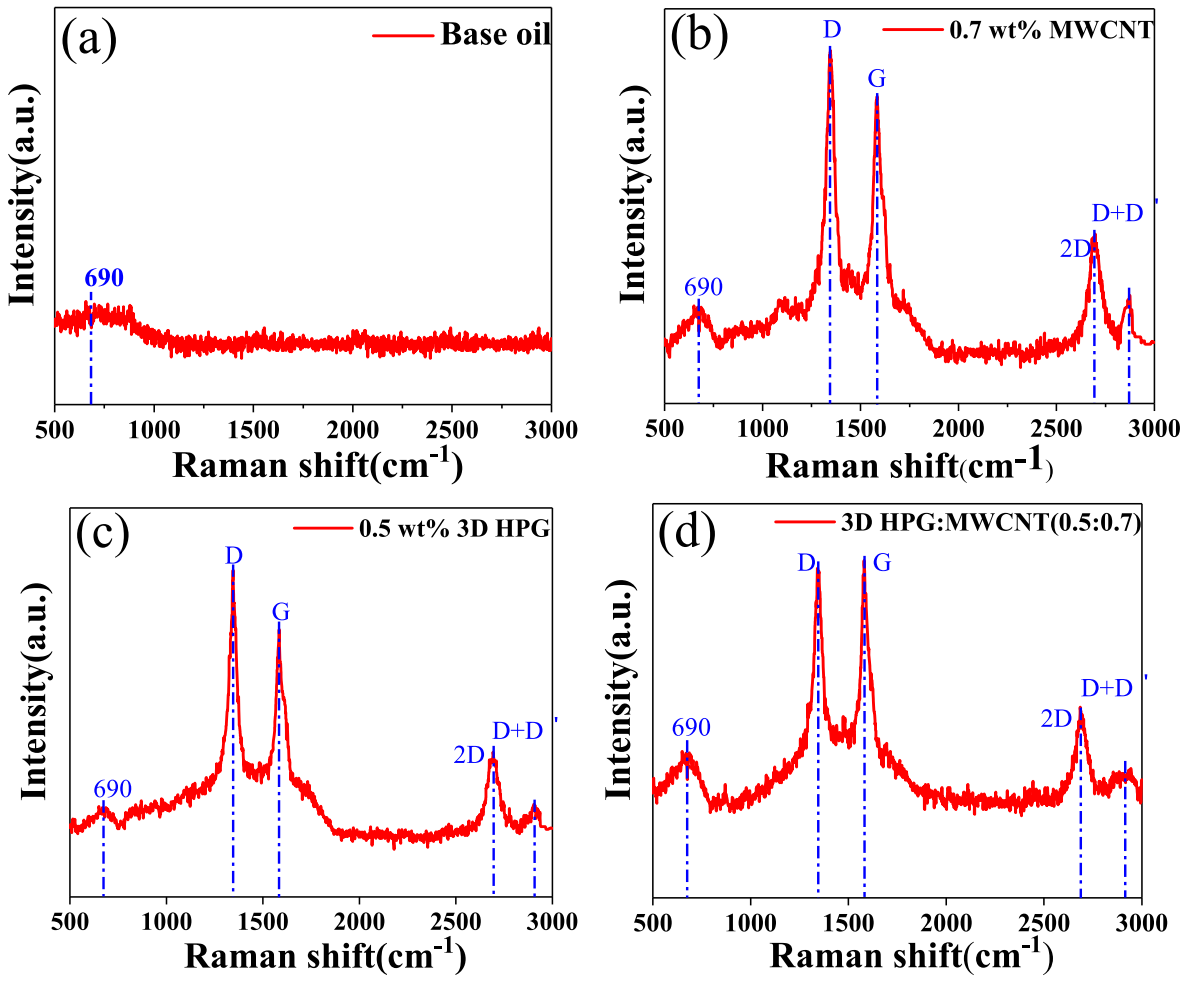


Fig. 9. Raman spectra of worn surface lubricated by different nanofluids under 4 Hz - 10 N conditions for 30 min. (a) at base oil case; (b) at 0.7 wt% MWCNT nanofluids case; (c) at 0.5 wt% 3D HPG nanofluids case; (d) at 3D HPG:MWCNT (0.5:0.7) combination nanofluids case.

From Fig. 12(a), when pure 3D HPG nanoparticle is involved in lubrication, the 3D HPG nanoparticle can easily enter and adsorb into the metal friction interface to form a parallel lubricating protective film, due to their good wettability compared to MWCNT nanoparticle (in Fig. 11). During the friction process, the counter-abrasive motion of the friction pair is transformed into slipping between the 3D HPG layers in the lubricating protective film, which dramatically reduces friction and wear. Besides, under heavy load conditions, owing to extremely high shear forces, the lamellar structure of 3D HPG is torn into two dimensional graphene, which can still provide lubrication by adsorption and filling effects.

Fig. 12(b) illustrates the lubrication mechanism of MWCNT nanoparticle. Because of its the nanoscale size, MWCNT nanoparticle can enter the friction interface with the castor oil molecules and fill the furrows and pits of the contact surfaces, making the contact surface smooth. Simultaneously, because of its large axial size and tubular structure, the MWCNT nanoparticle can play the role of micro-lubricated bearings, which substantially reduces the friction coefficient and wear volume. When the experimental condition tends to heavy load, the structure of some of the MWCNT nanoparticles is fractured due to their radial and axial size difference, resulting in the difficulty of forming a dense lubricating film. However, MWCNT nanopar-ticles of well-preserved structure are still available for lubrication.

When 3D HPG:MWCNT (0.5:0.7) mixed nanofluids is used for lubrication (shown in Fig. 12(c)), the MWCNT nanoparticles in hybrid nanofluids fill nano scale furrows and pits on friction pair surfaces to make the surfaces smooth, which makes it easier for suspended 3D HPG nanoparticles to adsorb onto relatively smooth surfaces to form a dense and uniform protective film, reducing the local contact pressure. Therefore, with the help of MWCNT, 3D HPG can effectively reduce friction and wear. Furthermore, in the case of heavy load, with the gradual decrease of film thickness and viscosity, the individual suspended 3D HPG or MWCNT nanoparticles are susceptible to be squeezed out of the friction contact zone under high contact pressure. In contrast, the suspended 3D HPG/HWCNT mixed nanoparticles can be fully adsorbed on the friction surface to avoid direct contact between tribo-pairs owing to its best wettability (based on Fig. 11(d)). Under the sustained action of shear force, 3D HPG is torn into 2D nanosheets due to its easy shear property. In this way, a sandwich composite lubrication structure of 2D graphene nanosheets/MWCNT/2D graphene nanosheets is formed between the friction interfaces: the outer 2D graphene nanosheets continue to maintain a relatively smooth surface by filling and adsorption, and the MWCNT nanoparticles in the middle of the 2D graphene nanosheets has a rolling effect, which is enable to further improve the friction-reduction and anti-wear properties. Because of this special rolled bearing-like synergistic lubrication effect between 3D HPG and MWCNT nanoparticles, the 3D HPG:MWCNT (0.5:0.7) shows the best tribological performance.

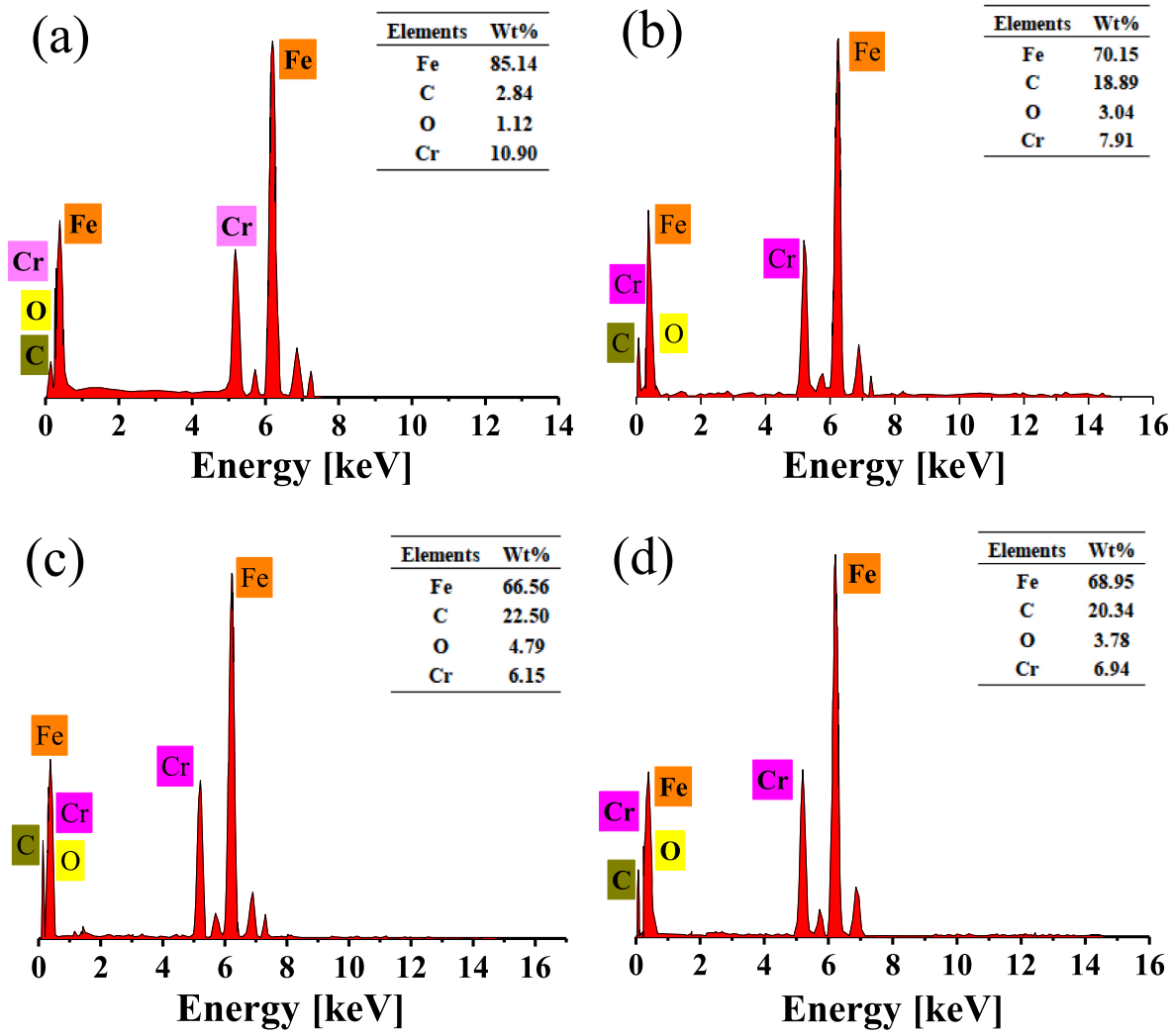


Fig. 10. EDS spectrum of worn surface lubricated by nanofluids under 4 Hz - 10 N conditions. (a) base oil; (b) 0.5 wt% 3D HPG; (c) 0.7 wt% MWCNT; (d) 3D HPG: MWCNT (0.5:0.7).

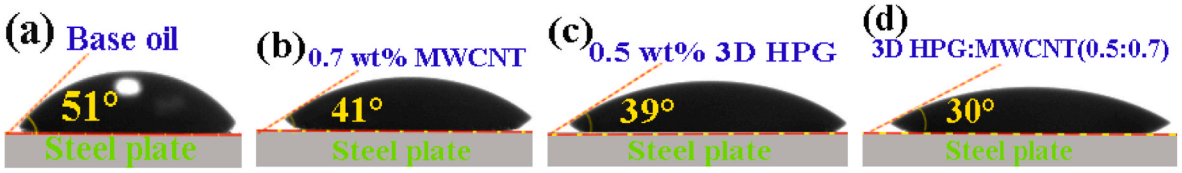


Fig. 11. Measured contact angle of four different oil-based lubricants on the surface of 304 stainless steel alloy. (a) base oil; (b) 0.7 wt% MWCNT nanofluids; (c) 0.5 wt% 3D HPG nanofluids; (d) 3D HPG:MWCNT (0.5:0.7) combination nanofluids.

4. Conclusion

In this study, through a series of experiments, the optimal concentration ratios of individual nanoparticle (3D HPG or MWCNT) and 3D HPG/MWCNT mixed nanoparticle are successively obtained. On this basis, under heavy load conditions, the tribological characteristics of 3D HPG, MWCNT and 3D HPG/MWCNT hybrid nanoparticles are further investigated for comparison. Due to the consistency of the steel plate and ball wear analysis, by using a series of characterization results (SEM images, Raman spectroscopy, EDS data, and contact angle measurements) on steel plates, the specific conclusions can be stated as follows:

(1) For the castor oil containing MWCNT nanoparticle, the optimal concentration of nano MWCNT is 0.7 wt%. Under 4 Hz - 10 N optimal working condition, compared to castor oil, the friction coefficient, wear volume, abrasion width and abrasion depth are reduced by 13%, 28%, 18% and 16%, respectively.

(2) For castor oil incorporating 3D HPG nanoparticle, its optimized concentration is 0.5 wt%. Under 4 Hz - 10 N optimal working condition, the friction coefficient, wear volume, abrasion width and abrasion depth of nano 3D HPG are reduced by 16%, 36%, 26% and 30%, separately.

(3) In terms of castor oil containing 3D HPG/MWCNT hybrid nanoparticle, the optimal concentration ratio of 3D HPG to MWCNT is

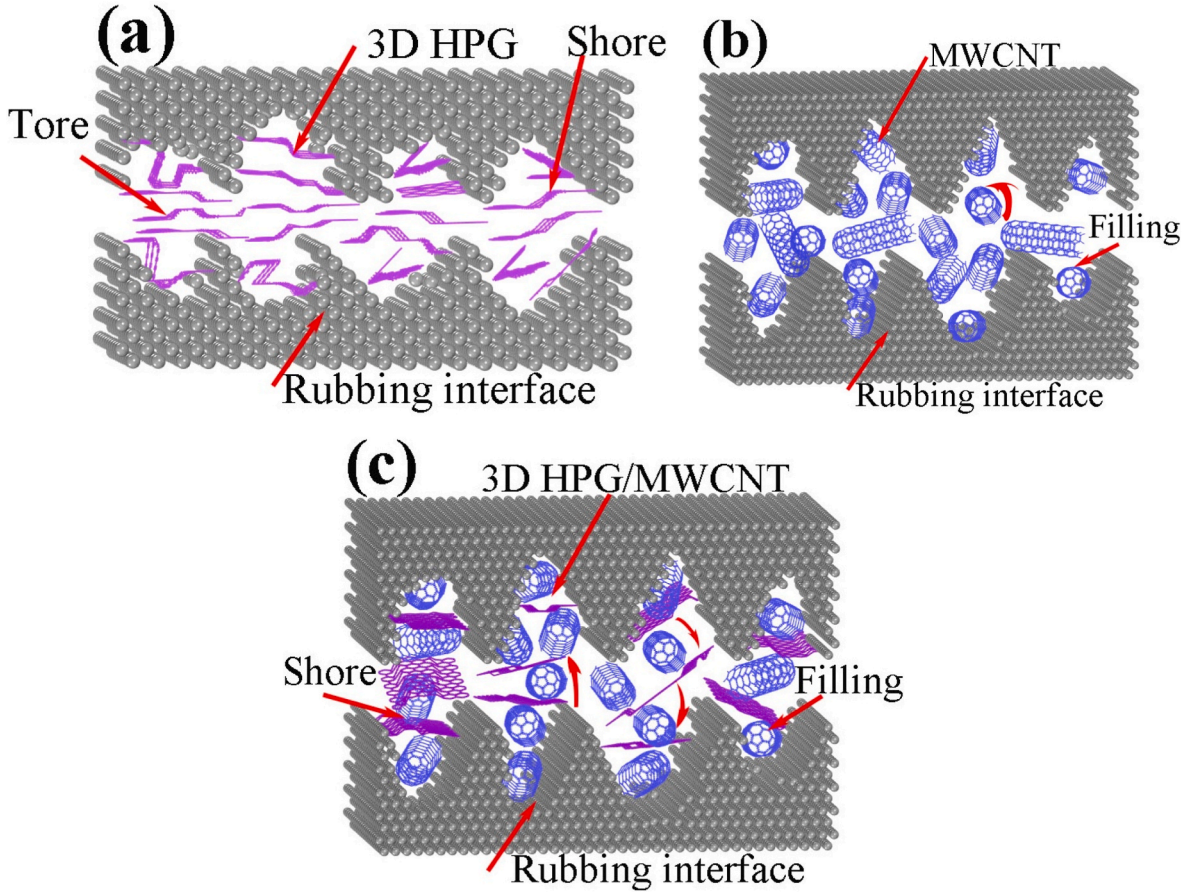


Fig. 12. Diagram of lubrication mechanism. (a) 0.5 wt% 3D HPG nanofluids; (b) 0.7 wt% MWCNT nanofluids; (c) synergistic lubricating effects of 3D HPG:MWCNT (0.5:0.7) combination nanofluids.

0.5:0.7. Compared to castor oil, the friction coefficient, wear volume, wear width and wear depth of 3D HPG:MWCNT (0.5:0.7) hybrid nanoparticle are decreased by 26%, 42%, 39% and 35% at 4 Hz - 10 N condition, respectively. Especially for the 20 N case, the wear loss can be reduced by 48%.

(4) Compared with 3D HPG or MWCNT nanoparticle alone, 3D HPG/MWCNT mixed nanoparticle shows more stable and excellent lubrication properties. Moreover, with the increase of load, the effect of friction-reduction and anti-wear is more obvious, which originates from the synergistic lubrication performance of the hybrid nanoparticles.

(5) The mechanism of synergistic lubrication is as follows: for the MWCNT nanoparticle, it can make the friction surfaces smoother by filling grooves and pits, which makes suspended 3D HPG is more easily adsorbed at the friction interface to form a protective film. Due to its easy shear properties, the 3D HPG is decomposed into a 2D graphene lamellar structure, in turn, the rolled bearing-like synergistic effect of a 2D graphene nanosheets/MWCNT/2D graphene nanosheets composite structure dramatically reduces the friction and wear of tribo-pairs surfaces.

Therefore, 3D HPG/MWCNT hybrid nanoparticle, as castor oil additives, has potential application in the field of tribology, especially for the lubrication of machine parts under heavy load conditions, thanks to their environmental friendliness and excellent tribological properties.

Credit author statement

Tiancheng Ouyang: Formulation of overarching research goals and experimental scheme. Wentao Tang: Experimental test, data analysis and English writing. Mingming Pan: Data collation and material characterization. Jing Tang: Content modification. Haozhong Huang: English revision.

Declaration of interest

None.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Abbreviations	
3D HPG	3D hierarchical porous graphene
MWCNT	multi-walled carbon nanotube
IF-WS_2	fullerene-like tungsten disulfide
ZnO	zinc oxide
SiO_2	silicon dioxide
TiO_2	titanium dioxide
CT80	coiled tubing steel
PAO	poly alpha olefin
T161	high molecular weight polyisobutylene succinimide
316 ASS	austenitic stainless stee
KOH	potassium hydroxide
SEM	scanning electron microscope
EDS	energy dispersive spectrometer

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